HARIKRISHNAN K S

Electronics & Communication Engineering Undergraduate | Analog VLSI | Analog IC Design | FPGA
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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with a focus on Analog IC Design and Analog VLSI. Familiar with analog design flows, semiconductor devices, PDKs, circuit simulation, and physical verification concepts. Currently developing skills in gm/ID methodology, CMOS bandgap reference design, and open-source EDA tools, including Xschem, Magic, and ngspice. Also experienced with FPGA design and Verilog HDL, with an interest in entry-level analog circuit design roles.

EDUCATION

B.Tech, Electronics & Communication Engineering *Expected 2027*
Government Engineering College, Sreekrishnapuram, Palakkad
APJ Abdul Kalam Technological University, Kerala
CGPA : **8.07** *(up to semester 6)*

TECHNICAL SKILLS

Analog IC Design:	Analog IC Design Fundamentals, Semiconductor Devices, gm/ID Methodology, Bandgap Reference Fundamentals, Filter Design, Analog Layout Basics
PDKs & EDA Tools:	SKY130, GF180MCU, Xschem, ngspice, Klayout, LTspice
Physical Verification:	DRC, LVS, Parasitic Extraction (PEX) - Concepts
Digital Design & FPGA:	Verilog HDL, RTL Design, FSM, Vivado 2022.2, Icarus Verilog, GTKWave, Zynq-7000
Programming:	C, C++, Python
EmbeddedSystems:	STM32, ESP32, Arduino, UART, I2C, SPI

TECHNICAL PROJECTS

Design and Layout of CMOS Bandgap Reference Circuit | SKY130 PDK • Xschem • Klayout • Ngspice

- Designed and simulated a CMOS Bandgap Reference (BGR) circuit using the SKY130 130-nm CMOS PDK, combining PTAT and CTAT voltage generation for temperature-compensated reference voltage generation.
- Implemented a five-transistor OTA, voltage-output stage, and voltage-comparison startup topology, applying gm/ID-based transistor sizing and custom IC layout concepts with DRC/LVS verification.

Adaptive Real-Time Traffic Light Control Using FPGA | Zedboard • FSM • Verilog • Vivado

- Developed FPGA-based traffic control system in Verilog HDL, improving traffic flow by 25% on Zynq-7000 (ZedBoard) using FSM-based control logic.
- Simulated and verified system behavior using GTKWave and Icarus Verilog

Enviroguard – Environmental Monitoring System | STM32 • IoT • Proteus

- Designed STM32-based environmental monitoring system, improving power efficiency by 30% with real-time sensor alerts (latency <50 ms).
- Validated circuit design in Proteus prior to hardware implementation.

RFID-Based Attendance System | ESP8266 • IoT

- Developed RFID-based attendance system using ESP8266, enabling 100% contactless tracking and real-time data processing.

EXPERIENCE

Analog IC Design Scholar *Present*
CuriousIC Edu (Batch 3)

- gm/ID Design Methodology, gf180MCU - PDK, SKY180 - PDK, Filter design, BandGap Reference circuits.

Analog IC Design Trainee Intern *June 2026*
Sophrosyne Semiconductor, Kunjibettu, Udupi

- Gained foundational exposure to analog IC design flow, semiconductor devices, analog building blocks, layout techniques, and physical verification concepts (DRC/LVS/ERC/PEX).

WORKSHOPS & CERTIFICATIONS

Advanced Driver Assistance Systems & Machine Learning Workshop ***November 2023***
Indian Institute of Technology Delhi

Introduction to Internet of Things **November 2024**
NPTEL

ACHIEVEMENTS & LEADERSHIPS

- Active member, Robotics Club, GEC Sreekrishnapuram; Volunteer, IEEE Young Buddies 2.0; UI/UX Designer & Design Head, ECE Department (incl. department magazine).